PROJECT SYNOPSIS: SKILLBRIDGE AI

Bridging the Gap Between Higher Education and Career Readiness with AI

Project Domain: EdTech / AI-Driven Career Tech	**Target Audience:** Students, Job Seekers, Switchers
Framework: Next.js 15.3 (Turbopack) & React 18	**AI Engine:** Google Gemini 2.5 Flash & Genkit 1.20
Databases: MongoDB (Mongoose) & Firebase Firestore	**Authentication:** Firebase Auth (OAuth 2.0 & Email/Pass)

1. Executive Summary & Abstract

SkillBridge AI is an intelligent, multi-dimensional career readiness platform engineered to bridge the growing divide between academic university curricula and current tech industry expectations. Leveraging Generative AI (Google Gemini 2.5 Flash orchestrated via Google Genkit) alongside Next.js 15, the platform delivers autonomous skill gap diagnoses, dynamic 12-week learning roadmaps, automated ATS resume scoring (ResumeRadar), a 24/7 interactive career coach, and direct connectivity to 20,000+ industry mentors.

2. Problem Statement & Motivation

- **Academic-Market Skill Disparity:** University programs cannot rapidly update to match modern industry demands in cloud computing, TypeScript ecosystems, and AI toolchains.
- **The ATS 'Black Hole':** Over 70% of qualified resumes are filtered out by Applicant Tracking Systems due to keyword mismatches and incompatible formatting.
- **Static Learning Materials:** Generic video courses fail to adapt to a candidate's background, available hours, and explicit career milestones.
- **Mentorship Inaccessibility:** 1-on-1 industry guidance from senior engineers is typically expensive, fragmented, or inaccessible to young professionals.

3. Core Functional Modules

Module	Core Functionality	Technology Stack
■ **AI Career Coach**	Interactive 24/7 career mentor providing real-time technical guidance, role transition roadmaps, and mock interview prep.	Gemini 2.5 Flash, Genkit flows, Next.js API
■ **12-Week Roadmap Generator**	Constructs adaptive week-by-week study schedules with curated docs, coding assignments, and capstone projects.	Genkit AI, MongoDB, Dynamic Schema
■ **ResumeRadar (ATS Analyzer)**	Evaluates PDF/Text resumes against targeted job descriptions; calculates match %, ATS scoring, and bullet point rewrites.	Gemini NLP, Multi-part Parser, Radix UI
■ **Skill Gap & Trends**	Interactive charts highlighting missing competencies against current high-demand engineering positions.	Recharts Data Viz, Firestore Sync
■ **Mentor Connect Network**	Searchable directory of 20,000+ verified mentors across top tech enterprises (Google, Meta, Amazon, Microsoft).	Firebase Auth, Firestore, REST APIs

4. Technical Architecture & Component Stack

SkillBridge AI is constructed using a high-throughput, modular, serverless architecture structured across four distinct layers:

- **Frontend & Client Tier:** Next.js 15.3 App Router with Turbopack bundler, React 18, Tailwind CSS, Radix UI headless components, Lucide Icons, and Framer Motion micro-interactions.
- **Server & API Tier:** Serverless API endpoints on Next.js handling synchronous AI streaming, JSON schema validation via Zod, and connection-pooled database queries.
- **AI & Inference Tier:** Google Genkit 1.20 orchestration engine interfacing with Google Gemini 2.5 Flash for high-speed, deterministic structured JSON outputs with Omnidim AI backup fallback.
- **Data & Persistence Layer:** Dual-database setup combining MongoDB (via Mongoose) for structured learning roadmaps, course repositories, and chat archives, alongside Firebase Firestore for real-time authentication and user metadata.

5. Database Strategy & Data Security

- **Multi-Database Synergy:** MongoDB provides document flexibility for complex weekly curriculum trees, while Firebase Firestore enables millisecond-level syncing for live user sessions and mentor availability.
- **Robust Authentication:** Firebase Authentication with support for email/password and OAuth 2.0 (Google Login) with JWT-validated Next.js route middleware.
- **Input Sanitization & Schema Validation:** Strict client and server-side validation using Zod prevents prompt injection and guarantees structured outputs from LLM flows.
- **Server Polyfill Protection:** Custom polyfills guarantee reliable SSR execution across diverse Node runtimes without memory leaks or storage conflicts.

6. Target Beneficiaries & Market Impact

Stakeholder	Direct Value Proposition
College Students & Freshers	Gain clarity on exact tech stacks demanded by recruiters, receive personalized weekly study roadmaps, and practice technical interviews.
Active Job Seekers	Instant ATS resume compatibility scoring, keyword optimization, and actionable feedback to pass automated HR filters.
Career Switchers	Deconstructs target roles (e.g. Data Science, Cloud Architect) into practical milestones tailored to existing skills.
Tech Mentors	Seamless platform to discover high-potential mentees and manage guidance sessions effectively.

7. Future Scope & Enhancements

- **Real-Time AI Voice Mock Interviews:** Conversational audio interviews simulating real hiring panel questions with instant speech analytics.
- **GitHub & Portfolio Intelligence:** Automated skill verification via repository commit analysis and code quality benchmarking.
- **Institutional Campus Placement Portal:** Enterprise analytics dashboards for college placement cells to monitor student readiness and batch performance in real-time.